Assignment 10.2 – Frappe Workspace Exploration

1. Objective

The objective of this assignment is to explore the Frappe Desk, understand existing DocTypes, study form structure and database schema, and create a simple custom Student DocType.

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2. Frappe Desk

Frappe Desk is the main user interface of the Frappe Framework.

It provides access to:

- Modules
- DocTypes
- Forms
- Reports
- List views
- Workspaces
- User management
- System settings

After logging into the development site, the Frappe Desk was explored to understand its different sections and features.

Screenshot: Insert Frappe Desk screenshot here.

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3. Existing DocType Exploration

A DocType in Frappe defines the structure and behavior of a particular type of document.

Examples of existing DocTypes include:

- User
- Role
- ToDo
- File
- Website Settings

The **User DocType** was explored to understand how fields are organized.

Some fields available in the User DocType include:

- Email
- First Name

- Last Name
- Enabled
- Language
- Time Zone

The DocType provides the form structure used to enter and manage information.

Screenshot: Insert existing DocType screenshot here.

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4. Form Structure

A Frappe form is generated from the corresponding DocType.

A form generally contains:

- Document name
- Fields
- Field labels
- Field types
- Mandatory fields
- Save button
- Actions
- Document information

Different field types can be used depending on the type of information being stored.

Examples:

Field Type	Purpose
Data	Stores text
Email	Stores email address
Phone	Stores phone number
Int	Stores integer values
Select	Provides predefined options
Date	Stores dates
Check	Stores Yes/No values

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5. Database Schema

Frappe uses a database to store DocType records.

For example, a DocType named Student generally has a corresponding database table named:

tabStudent

Frappe also maintains standard document fields such as:

- name
- owner
- creation
- modified
- modified_by
- docstatus
- idx

The database was explored using the MariaDB command available through Bench.

Command:

cd ~/frappe-bench

bench --site development.localhost mariadb

To display tables:

SHOW TABLES;

To check the Student table:

SHOW TABLES LIKE 'tabStudent';

To view Student records:

SELECT * FROM tabStudent;

To exit MariaDB:

exit;

Note: Actual passwords are not included in this documentation.

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6. Creating Student DocType

A custom DocType named **Student** was created in Frappe Desk.

The following fields were added:

Field Label	Field Type	Purpose
Student Name	Data	Stores student name

Email	Email	Stores email address
Phone	Phone	Stores phone number
Age	Int	Stores student's age
Course	Select	Stores selected course

For the **Course** field, the following options were added:

Java

Python

React

Data Science

Other

The Student DocType was saved successfully.

Screenshot: Insert Student DocType configuration screenshot here.

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7. Student Record

A sample Student record was created.

Student Details

Student Name: Radhika Rawol
Email: radhika@example.com
Phone: 9876543210
Age: 24
Course: Java

The record was saved successfully.

Screenshot: Insert Student record screenshot here.

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8. Database Verification

After creating the Student record, the database was checked using:

bench --site development.localhost mariadb

The Student table was verified using:

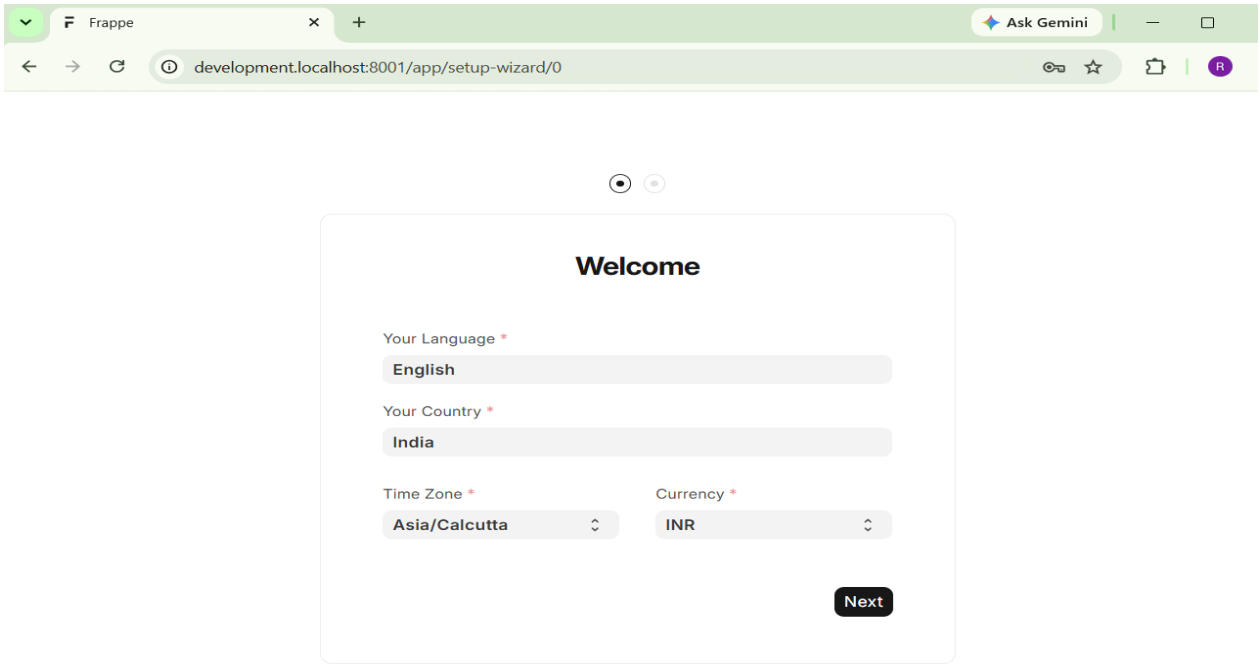
SHOW TABLES LIKE 'tabStudent';

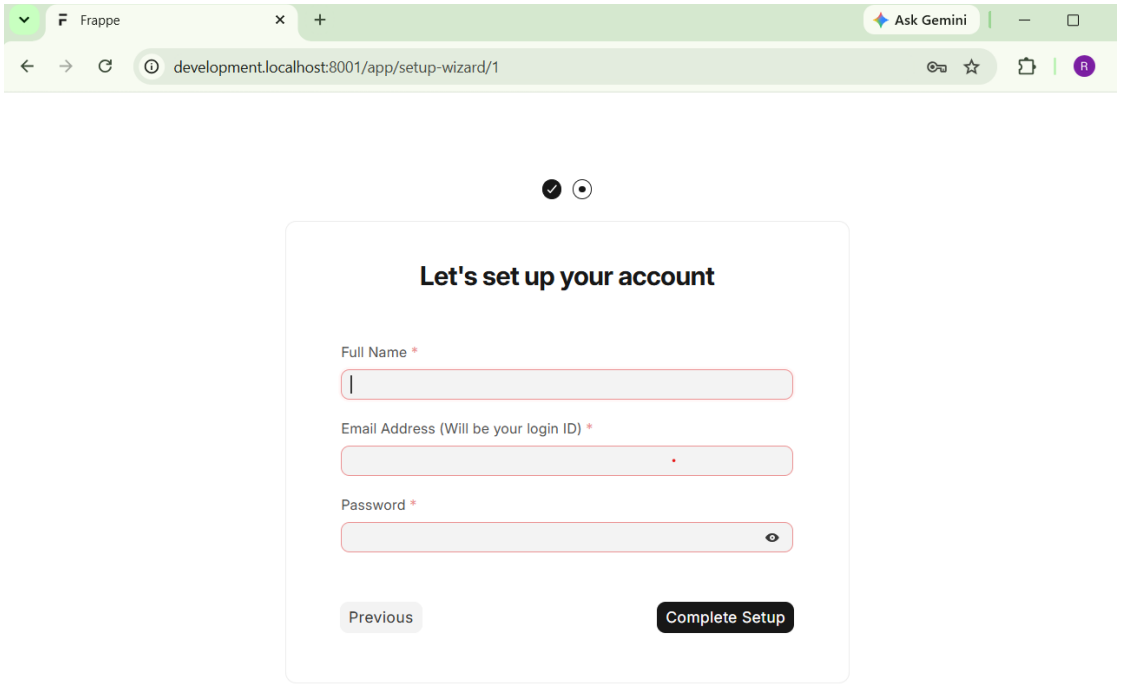
The Student records were checked using:

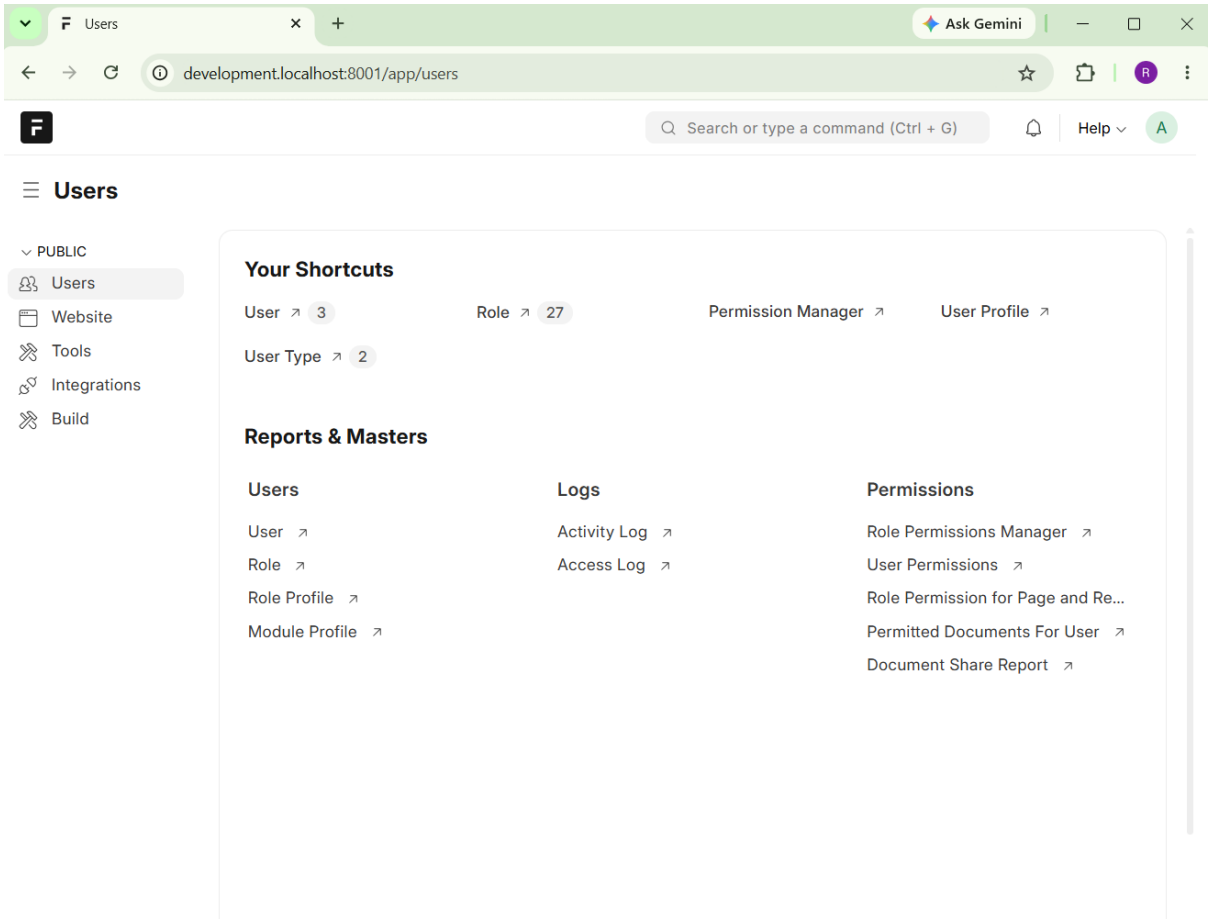
SELECT * FROM tabStudent;

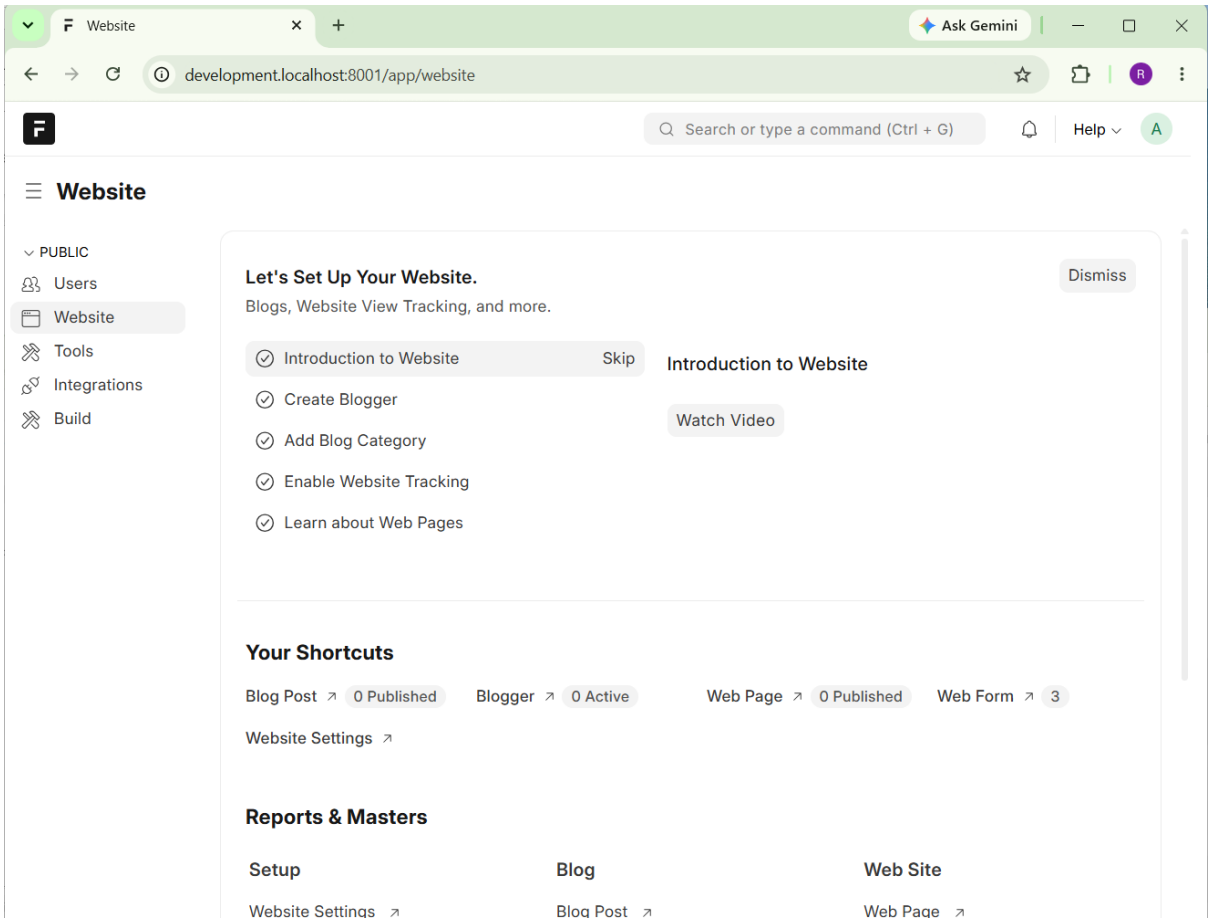
This confirmed that the Student DocType was connected to the database and that the created record was stored successfully.

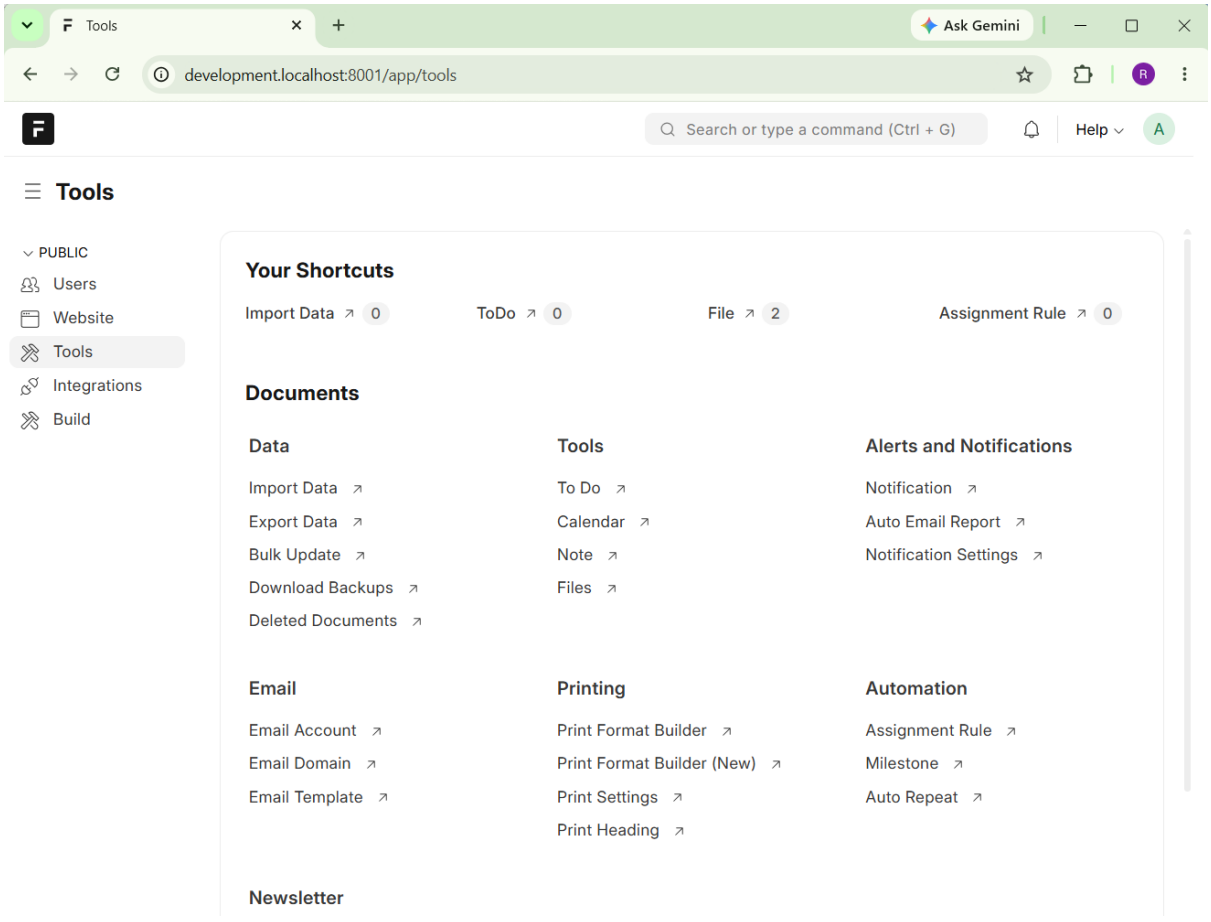
Screenshots – Assignment 10.2

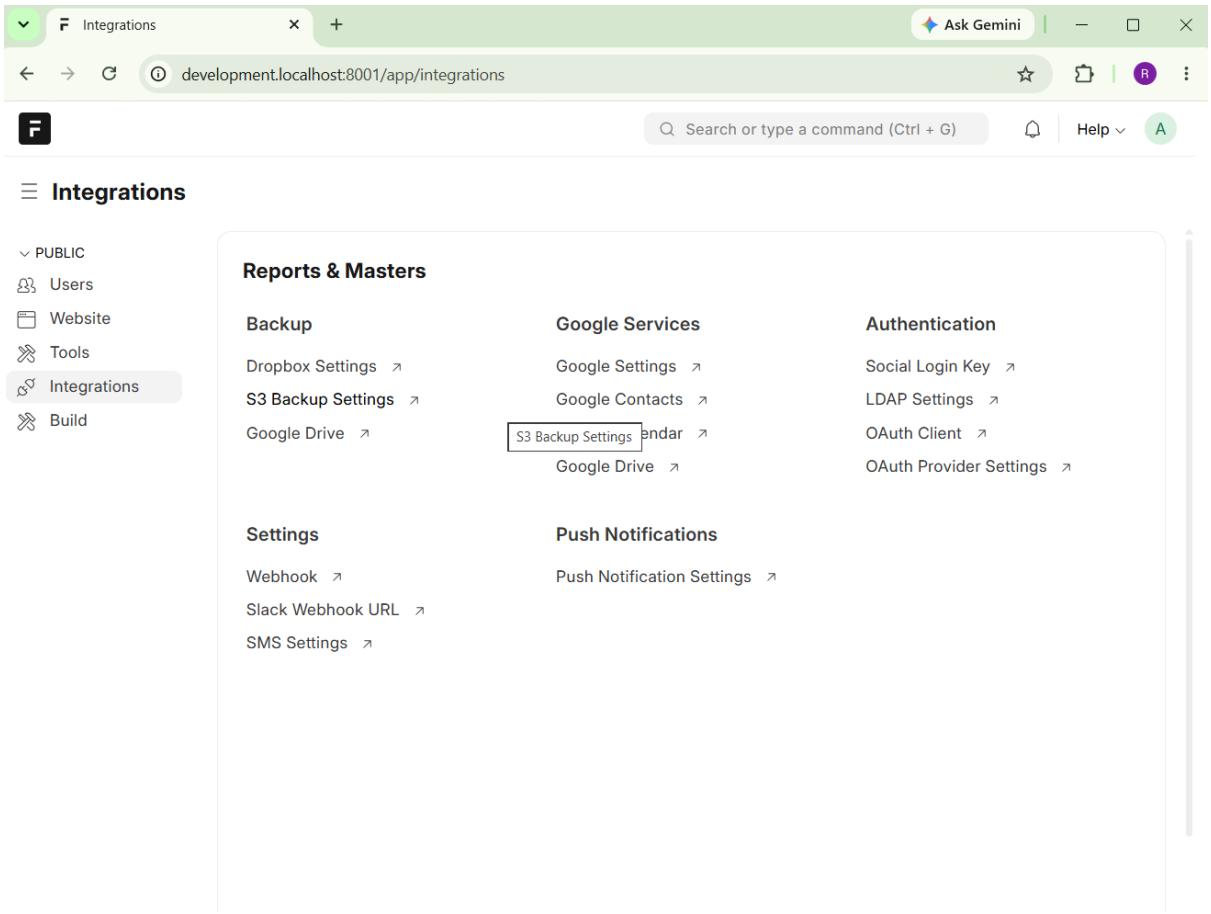


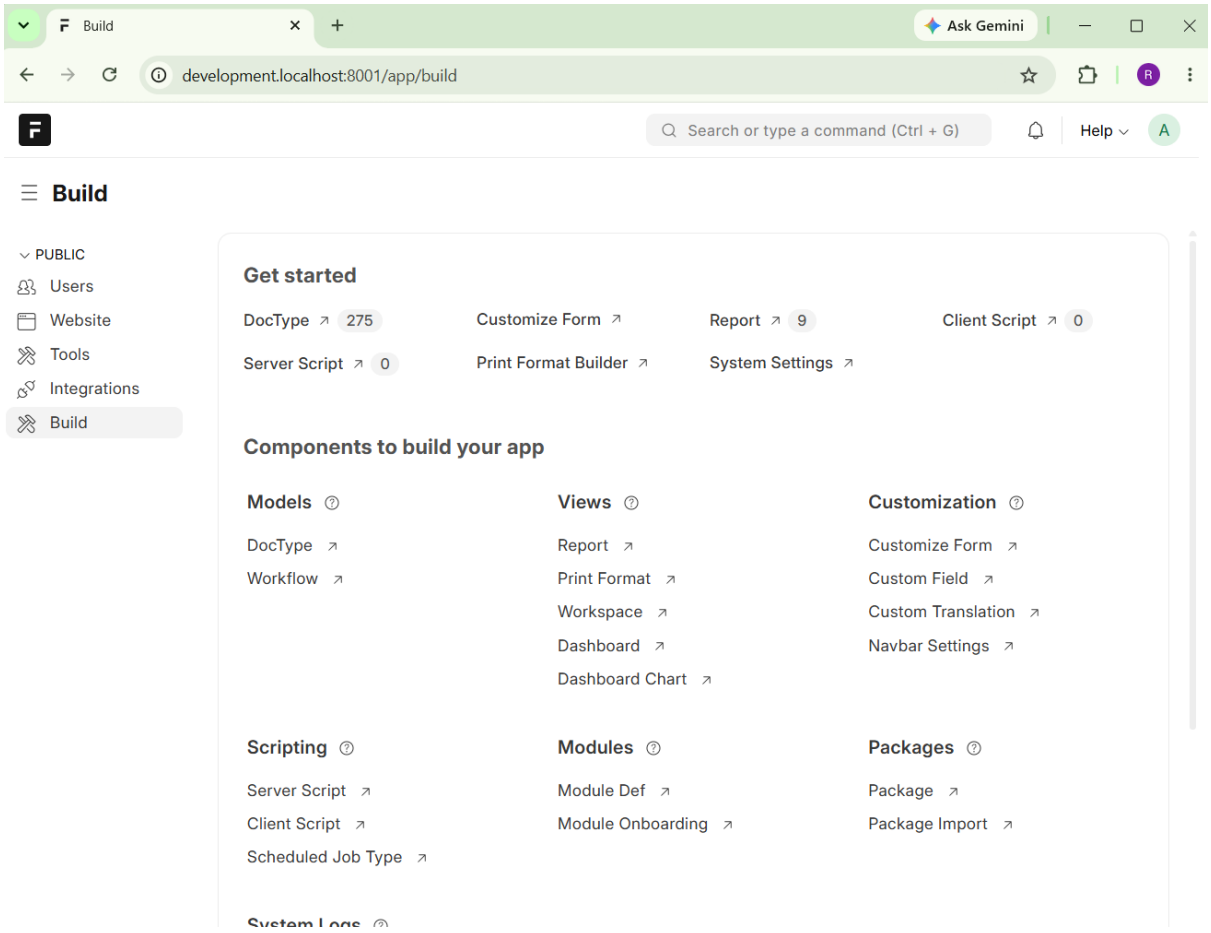



Users
Ask Gemini
development.localhost:8001/app/users
Search or type a command (Ctrl + G)
Help
A
Users
PUBLIC
Users
Website
Tools
Integrations
Build
Your Shortcuts
User 3
Role 27
Permission Manager
User Profile
User Type 2
Reports & Masters
Users
User
Role
Role Profile
Module Profile
Logs
Activity Log
Access Log
Permissions
Role Permissions Manager
User Permissions
Role Permission for Page and Re...
Permitted Documents For User
Document Share Report


Website
Ask Gemini
development.localhost:8001/app/website
Search or type a command (Ctrl + G)
Help
A
Website
PUBLIC
Users
Website
Tools
Integrations
Build
Let's Set Up Your Website.
Blogs, Website View Tracking, and more.
Dismiss
Introduction to Website Skip
Create Blogger
Add Blog Category
Enable Website Tracking
Learn about Web Pages
Introduction to Website
Watch Video
Your Shortcuts
Blog Post 0 Published
Blogger 0 Active
Web Page 0 Published
Web Form 3
Website Settings
Reports & Masters
Setup
Website Settings
Blog
Blog Post
Web Site
Web Page


Tools
Ask Gemini
development.localhost:8001/app/tools
Search or type a command (Ctrl + G)
Help
A
Tools
PUBLIC
Users
Website
Tools
Integrations
Build
Your Shortcuts
Import Data 0
ToDo 0
File 2
Assignment Rule 0
Documents
Data
Import Data
Export Data
Bulk Update
Download Backups
Deleted Documents
Tools
To Do
Calendar
Note
Files
Alerts and Notifications
Notification
Auto Email Report
Notification Settings
Email
Email Account
Email Domain
Email Template
Printing
Print Format Builder
Print Format Builder (New)
Print Settings
Print Heading
Automation
Assignment Rule
Milestone
Auto Repeat
Newsletter


Integrations
Ask Gemini
development.localhost:8001/app/integrations
Search or type a command (Ctrl + G)
Help
A
Integrations
PUBLIC
Users
Website
Tools
Integrations
Build
Reports & Masters
Backup
Dropbox Settings
S3 Backup Settings
Google Drive
Google Services
Google Settings
Google Contacts
S3 Backup Settings
Google Drive
Authentication
Social Login Key
LDAP Settings
OAuth Client
OAuth Provider Settings
Settings
Webhook
Slack Webhook URL
SMS Settings
Push Notifications
Push Notification Settings



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9. Testing

The following tests were performed successfully:

Test	Result
Ubuntu environment checked	Successful
Python installed	Successful
MariaDB installed	Successful
Redis installed	Successful
Node.js installed	Successful
Yarn installed	Successful
Frappe Bench installed	Successful
Frappe Bench created	Successful
Development site created	Successful

Frappe Desk opened	Successful
Existing DocType explored	Successful
Student DocType created	Successful
Student record created	Successful
Database table verified	Successful
Database record verified	Successful

End of Assignment

Submitted by:
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